

Leopardxv

GONGCHUANG

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README

Linux Learning System

AI-Enhanced Linux Online Learning Platform. Students practice Linux commands in a real terminal (Docker container) with AI-powered real-time analysis and learning guidance.

Architecture

Frontend (Next.js 15)

xterm.js Terminal ↔ WebSocket ↔ FastAPI

AI Learning CardsREST API

Backend (FastAPI)

Auth (JWT Cookie) · Terminal (Docker) · AI (DeepSeek) |

PostgreSQL

Docker Daemon

DeepSeek API

Quick Start

Prerequisites

- Python 3.12+
- Node.js 20+
- Docker & Docker Compose
- PostgreSQL 16
- DeepSeek API key (platform.deepseek.com)

1. Clone & Setup Backend

```
git clone https://github.com/Leopardxv/GONGCHUANG.git
cd linux-learning-system/backend

# Create virtual environment
python3 -m venv .venv
source .venv/bin/activate

# Install dependencies
pip install -r requirements.txt

# Configure environment
cp .env.example .env
# Edit .env: set LL_JWT_SECRET and LL_AI_API_KEY

# Start database
docker compose -f ../docker/docker-compose.yml up -d

# Create tables
python -c "
import asyncio
from app.database import engine
from app.models.base import Base
from app.modules.auth.model import User
from app.modules.terminal.model import DockerInstance, TerminalLog
from app.modules.ai.model import AIAnalysisLog
async def init():
    async with engine.begin() as conn:
        await conn.run_sync(Base.metadata.create_all)
asyncio.run(init())
"

# Start backend
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

2. Setup Frontend

```
cd ../frontend

# Install dependencies
npm install

# Start dev server
npm run dev
```

3. Build Student Docker Image

```
cd ..
docker network create ll-student-net
bash scripts/build_student_image.sh
```

4. Open Browser

- Home: `http://localhost:3000`
- Swagger: `http://localhost:8000/docs`

Project Structure

```
linux-learning-system/
├── frontend/           # Next.js 15 App Router
│   ├── app/           # Pages & layouts
│   ├── components/    # TerminalPanel, AuthProvider
│   ├── hooks/         # useTerminal, useAuth
│   ├── services/      # API client, auth, terminal, AI
│   ├── stores/        # Zustand stores
│   └── config/        # API, layout, feature flags
├── backend/           # FastAPI
│   ├── app/           # Main app, config, DB, middleware
│   ├── modules/       # auth, terminal, ai
│   ├── prompts/       # AI prompt templates
│   └── tests/
├── docker/            # docker-compose, student Dockerfile
├── scripts/           # Build scripts
└── docs/              # Architecture & API docs
```

API Endpoints

Method	Path	Description
POST	<code>/api/v1/auth/register</code>	Register
POST	<code>/api/v1/auth/login</code>	Login (JWT cookie)
POST	<code>/api/v1/auth/logout</code>	Logout
GET	<code>/api/v1/auth/me</code>	Current user
GET	<code>/api/v1/auth/stats</code>	User learning stats
WS	<code>/api/v1/terminal/connect</code>	Terminal WebSocket
POST	<code>/api/v1/ai/analyze-command</code>	AI command analysis
POST	<code>/api/v1/ai/chat</code>	AI chat
GET	<code>/health</code>	Health check

Environment Variables

Copy `backend/.env.example` to `backend/.env` and configure:

Variable	Description
<code>LL_DATABASE_URL</code>	PostgreSQL connection string
<code>LL_JWT_SECRET</code>	Random secret for JWT signing
<code>LL_AI_API_KEY</code>	DeepSeek API key
<code>LL_CORS_ORIGINS</code>	Allowed CORS origins (JSON array)
<code>LL_DOCKER_IMAGE</code>	Student Docker image name (default: <code>linux-student:latest</code>)

License

MIT

About

AI-Enhanced Linux Online Learning Platform - practice Linux commands in a real Docker terminal with AI-powered real-time analysis and learning guidance.

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